

Introductory Econometrics with Recitation: TA Session Week 8

Danny Po-Hsien Kang (康柏賢)

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Today's agenda

- 1 R Markdown
 - Introduction to R Markdown
 - Text Formatting
 - R Code Chunks
 - Chunk Options
 - Knit
 - Open a new R Markdown

Midterm

- Midterm: 9:30 AM - 12:00 PM, April 22 (Wed)
- You can bring: One cheat sheet, Calculator
- No questions about R will appear in the midterm exam!
- No TA Session in midterm week.
 - TA office hour: April 21 (Tue) 13:20-17:20

What is R Markdown?

- R Markdown allows you to run your R code and write your report in the same document.
- You can use it for:
 - assignments
 - quizzes
 - research reports
 - data analysis presentations
- Why Use R Markdown?
 - It keeps your code, results and explanation together.
 - It makes your analysis easier to read and reproduce.
 - If you want a clean and organized workflow, R Markdown is a very good choice.

When to Use R Markdown vs. R Script

- **Use R Markdown when you want to:**
 - submit homework or quizzes
 - combine text, code, tables, and plots
 - create a clean and reproducible report
- **Use an R script when you want to:**
 - write and test code quickly
 - clean data or build functions
- A common workflow is:
 - first write and test code in an `.R` script
 - then present the final analysis in an `.Rmd` file

Three Main Parts of an R Markdown File

- An R Markdown file is made up of three main parts:
 - YAML header
 - Markdown
 - R code chunks
- **YAML header:** basic information such as title, author, and output format
- **Markdown:** the part where you write text and format notes
- **R code chunks:** the part where you run R code and show outputs

YAML Header

- The YAML header appears at the very top of the .Rmd file.
- It is surrounded by ---
- It tells R Markdown what kind of document to create.
- **Example:**

```
---  
title: "My First R Markdown File"  
author: "Danny"  
output: html_document  
---
```

- You can change the output to pdf_document, word_document, or html_document.

Markdown: Text Formatting

- In R Markdown, you write your text using Markdown syntax.
- Markdown is a simple way to format text with plain symbols.
- If you have used tools such as:
 - Notion
 - HackMD
 - Typora

then you may already be familiar with Markdown.

Markdown: Headings, Bold, and Italics

- **Headings:**

```
# Heading 1 (Big Title)
## Heading 2 (Section)
### Heading 3 (Subsection)
```

- **Bold and Italics:**

```
*Italic text*
**Bold text**
```

- Use one `*` for italic and two `**` for bold.

Markdown: Lists and Links

• Lists and numbers:

```
- Item 1
- Item 2
  - Sub-item 2.1

1. First item
2. Second item
  1. Sub-item 2.1
```

• Links:

```
[Link to Google](https://www.google.com)
```

- Markdown is useful when you want to organize your explanation clearly.

Code Block: R Chunks

- You can insert R code using **code chunks**.
- We always surround the code with three backticks and start with {r}.
- To execute a code chunk, you can press the Run button in RStudio.
- **Syntax:**

```
```${r}  
library(here)
birth <- read.csv(here("data", "ncbirths.csv"))
summary(birth$weight)
```
```

- R code chunks in an Rmd file are evaluated relative to the directory that contains it, not the project folder.
- `here()` builds an absolute file path relative to the project root.

Code Block: Plot Example

- **Plot example:**

```
```{r, echo=FALSE}  
ggplot(birth, aes(x = weeks, y = weight)) +
 geom_point(color = "steelblue", size = 2.5) +
 labs(
 x = "Birth Weeks",
 y = "Baby Weight (pounds)"
)
```
```

- Here, `echo = FALSE` means the code will be hidden.
- Only the plot will appear in the final document.

Chunk Options

- You can control how each chunk behaves by adding **chunk options**.
- Common options:

```
echo = TRUE      # show code
echo = FALSE     # hide code
message = FALSE  # hide messages
warning = FALSE  # hide warnings
eval = FALSE     # show code but do not run it
```

- These options are useful when you want your output to be cleaner.

Knit: Output

- Once your R Markdown file is ready, click the **Knit** button at the top of RStudio.
- When you knit:
 - R runs all the code chunks in order.
 - It combines your text, code, and outputs into one final document.
 - It creates a nicely formatted HTML, Word, or PDF file.
- This is why R Markdown is a useful tool for reproducible analysis.

Settings in a new R Markdown

- Open a new R Markdown by File -> New File -> R Markdown
- You'll see two default chunks after opening it

```
```${r} setup, include=FALSE}  
knitr::opts_chunk$set(echo = TRUE) # Set echo = TRUE globally
```
```

```
```${r} cars  
summary(cars)
```
```

- cars, setup is the chunk label.
- r is the language executed in the chunk.